

Stoquify Agent Runtime Phase 2A Release-Control Hardening Execution Report

Execution date: 2026-07-24 **Governing prompt:** docs/agents-runtime/STOQUIFY_INTERNAL_ACTIVATION_GATE_REMEDIATION_AND_CONTROLLED_PILOT_READINESS_EXECUTION_PROMPT_2026-07-23.md **Authoritative phased plan:** docs/agents-runtime/STOQUIFY_AGENT_RUNTIME_PRACTICAL_EXECUTION_PLAN_2026-07-22.md **Active phase:** Phase 2A, deterministic read-only Command Agent **Phase result:** Repository hardening complete for this slice; Phase 2 remains partially blocked **Final decision:** READY FOR REMEDIATION VERIFICATION **Internal activation:** BLOCKED and not attempted **Phase 3:** NOT AUTHORIZED

1. Executive Decision

This execution closed three material technical gaps in the Phase 2A release-control plane:

1. Approval and certification commands are now idempotent, replay-safe, and payload-bound.
2. Global agent, skill, and tool definitions are independent of tenant rollout state.
3. Blocking drift on an active tenant release now causes an automatic guarded suspension with immutable evidence.

The controlled PostgreSQL migration and persistence smoke passed. The enabled Command Agent passed authenticated desktop and mobile Playwright certification, remained `PILOT_CERTIFIED`, and retained `activatedAt: null`.

During final evidence review, the raw Playwright JSON reporter was found to retain its web-server environment block. That block can contain configured credentials. The unsafe local artifact was replaced, the runner was changed to create a bounded credential-free report, the raw report is now deleted before certification, a static gate enforces the behavior, and browser certification was rerun so the database certificate is bound to the sanitized report hash.

Stoquify is not ready for a real controlled internal pilot. Real approvals and owners, a deployed scheduler and managed identity or secret, production-like alert delivery and acknowledgement, clean-commit CI evidence, and the full negative/degradation browser matrix remain incomplete. The correct next state is remediation verification in a production-like non-production environment.

2. What Was Executed

2.1 Release-command idempotency

The approval and pilot-certification commands now require an idempotency key. Each command:

- calculates a stable request hash;
- executes in a serializable transaction;
- looks up the package-scoped idempotency key before changing state;
- returns the original record with `replayed: true` for an identical retry;
- rejects reuse with different content as `RELEASE_IDEMPOTENCY_CONFLICT`;
- handles a concurrent unique-key race without duplicating evidence;
- records the key and request hash in audit evidence.

Nullable database columns preserve compatibility with historical rows. New service contracts require the values.

2.2 Tenant rollout and global registry separation

The global definition registry now remains:

- Agent definition: ACTIVE
- Agent rollout mode: SHADOW
- Skill definition: ACTIVE
- Tool definition: ACTIVE

Tenant pilot state is owned exclusively by `AgentActivationPackage`. Provisioning, activation, suspension, or rollback of one tenant can no longer change the global registry to `INTERNAL` or `PAUSED`.

This removes cross-tenant coupling and makes the release package the authoritative tenant rollout control.

2.3 Automatic drift suspension

The control-plane reconciler now evaluates release blockers independently from finding presentation. If an `ACTIVE_INTERNAL` package develops a blocking condition, it calls the canonical suspension command.

The suspension:

- is tenant-scoped;
- is concurrency-safe;
- is non-destructive;
- records a bounded reason;
- writes immutable audit evidence;
- leaves the global registry `ACTIVE/SHADOW`;
- reports the suspension count in reconciliation output.

2.4 Credential-free browser evidence

The browser runner now:

1. writes Playwright's raw JSON to a temporary file under `test-results`;
2. extracts only version, project identity, test identity, status, duration, retry, and aggregate statistics;
3. fails if forbidden environment or credential keys survive;
4. writes the bounded durable artifact;
5. deletes the raw report;
6. records certification only after sanitization.

The rerun produced a 2.8 KB artifact with zero forbidden-key matches. No raw reporter file remained.

3. Architecture and Ownership

The existing graph at `graphify-out/ordered-code-graph.json` was inspected. It contains the `daily-digest` route and `DailyHabitDigestDashboard` nodes, but it predates the agent-runtime work completed on 2026-07-22 through 2026-07-24. The current source tree was therefore treated as authoritative for new release-control dependencies.

The resulting ownership path is:

Daily digest UI -> protected agent action -> Command Agent service -> release authorization -> definition resolution -> read-only tool adapter -> evidence-backed output

Release governance follows:

protected release action -> canonical release-control service -> tenant-scoped PostgreSQL state -> append-oriented audit evidence

Operational assurance follows:

authenticated scheduler boundary -> control-plane reconciler -> Workflow Assurance check run/finding/incident -> alert delivery -> guarded suspension

No agent receives release-control permissions. No agent service gains direct ledger, filing, payment, inventory, payroll, close, approval, or permission-changing authority.

4. Files and Components Changed

Database

- prisma/schema.prisma
- prisma/migrations/20260724100000_agent_runtime_release_hardening/migration.sql

The migration adds nullable idempotency and request-hash columns plus package-scoped unique indexes for approval and certification commands. It is additive.

Runtime and governance services

- services/agents/agent-release-contracts.ts
- services/agents/agent-release-control.service.ts
- services/agents/agent-definition.service.ts
- services/agents/agent-control-plane-reconciliation.service.ts
- actions/agents/agent-release-control.actions.ts

Provisioning, certification, smoke, and gates

- scripts/agent-enabled-pilot-e2e-bootstrap.ts
- scripts/run-agent-enabled-pilot-e2e.js
- scripts/agent-release-control-postgres-smoke.ts
- scripts/agent-release-control-gate.js

Tests

- services/agents/__tests__/agent-release-control.service.test.ts
- services/agents/__tests__/agent-definition.service.test.ts
- services/agents/__tests__/agent-control-plane-reconciliation.service.test.ts

- existing agent and component suites

5. PostgreSQL Evidence

Migration

- Migrations discovered: 32
- Database state: up to date
- Migration safety checks: 8/8 ready
- Destructive SQL findings: 0
- Blockers: 0
- Secret values written to migration evidence: no

Final certified local package

- Package ID: cmryrcs790003ma10ivkyrim7
- Release version: e2e-ac30ee75314a-1784886529173
- Base commit recorded: ac30ee75314a0a2a2fcd6bd2ed65afb280aa0d5e
- Manifest hash: sha256:f6d5f1049e0baff64bb42503077f3fe801bd214bcf265289308f7db56f51c28a
- Certification ID: cmryrimlo0001map4fazfj3nm
- State: PILOT_CERTIFIED
- Activated at: null

The base commit does not include the dirty working-tree implementation. This makes the local certificate provisional until the same suite passes on a clean committed revision in CI.

Final browser evidence

- Playwright evidence SHA-256:
17ce50099a0ed64dc86c16b4ce41d10d047ed5f7210964995f217b033612272f
- Expected tests: 4
- Unexpected tests: 0
- Flaky tests: 0
- Forbidden evidence-key matches: 0
- Raw report retained: no

Final release-control smoke

- Package state: PILOT_CERTIFIED
- Product/security approvals: 2 from 2 distinct users
- Owner responsibilities: 6
- Certification records: 1
- Audit records: 18
- Approval replay: passed

- Certification replay: passed
- Conflicting payload: rejected as RELEASE_IDEMPOTENCY_CONFLICT
- Automatic suspension fixture: SUSPENDED
- Suspension audit: present
- Global registry after tenant suspension: ACTIVE/SHADOW
- Activation blocker: RELEASE_RECONCILIATION_STALE
- Alert transport: not ready because no webhook is configured

The smoke intentionally retains immutable assurance evidence. PostgreSQL correctly rejected an attempted cleanup of an immutable finding; the test was corrected to preserve the evidence instead of bypassing the control.

6. Verification Results

Verification	Result
npm run prisma:validate	Passed
npx prisma migrate status	Passed; 32 migrations, database current
npm run prisma:migration:safety:gate	Passed; 8/8, zero destructive risks
Controlled local migration deployment	Passed
npm run agent:release-control:postgres-smoke	Passed
Focused ESLint	Passed
npm run typecheck with 8 GB Node heap	Passed, no diagnostics
npx jest services/agents components/agents --runInBand	Passed; 18 suites, 54 tests
npm run agent:runtime:gates	Passed
Enabled-pilot desktop Playwright	Passed
Enabled-pilot mobile Playwright	Passed
Evidence redaction scan	Passed; zero forbidden-key matches
git diff --check	Passed; line-ending warnings only

The browser rerun emitted a Next.js development warning about future allowedDevOrigins behavior. It did not affect certification and should be addressed before relying on the same development-server topology long term.

7. Gate Matrix

Gate	Status	Evidence or blocker
Canonical release-control service	PASSED locally	Protected transitions and static gate
Direct activation bypass removal	PASSED	Provisioner remains draft/shadow only
Approval/certification idempotency	PASSED	Replay and conflict smoke assertions
Tenant/global rollout separation	PASSED	Global definitions remain ACTIVE/SHADOW
Blocking-drift suspension	PASSED	PostgreSQL suspension fixture and audit
Additive migration safety	PASSED	8/8 checks, no destructive findings
PostgreSQL persistence	PASSED locally	Final smoke ok: true
Credential-free Playwright evidence	PASSED	Distilled report, raw report deleted
Enabled desktop/mobile happy path	PASSED locally	4/4 including auth setup

Gate	Status	Evidence of blocker
Real product and security approval	BLOCKED	Authorized identities and decisions absent
Named operational owners	BLOCKED	Real roster, coverage, and acknowledgement absent
Five-minute deployed reconciliation	BLOCKED	Deployment adapter, schedule, and managed identity/secret absent
Production-like alert delivery	BLOCKED	Webhook, owner acknowledgement, and escalation proof absent
Clean-commit CI certification	BLOCKED	Current worktree is uncommitted
Negative/degradation browser matrix	PARTIAL	Unauthorized tenant/role, stale/partial/empty, timeout/retry/replay, accessibility, rollback, and DB non-mutation assertions remain
Retirement and long-term retention	PARTIAL	RETIRED state exists; dedicated command and retention operation remain

8. Security Finding and Required Response

The first local Playwright JSON artifact contained the web-server environment supplied by the reporter. It was untracked and was replaced before this report was published. The durable runner no longer retains that structure, and the rerun certificate is bound to the sanitized report.

Because credential values were present in a local evidence file, the security owner must rotate any non-test credentials that were loaded into that browser run before production-like certification. The repository does not prove whether local backup, indexing, or endpoint tooling copied the earlier artifact. Rotation is the conservative closure action.

No credential value is reproduced in this report.

9. Remaining Work to Unblock the Pilot

1. Commit the implementation and rerun all mandatory CI jobs from that exact commit.
2. Record real product and security approval using distinct authorized users.
3. Assign and acknowledge all six operational responsibilities with valid coverage and escalation references.
4. Deploy the five-minute reconciler with workload identity where available, otherwise a managed rotatable secret.
5. Configure production-like alert transport, deliver a test incident, obtain owner acknowledgement, and prove escalation.
6. Complete the negative and degradation Playwright matrix, including tenant isolation and prohibited-business-table non-mutation assertions.
7. Rotate any non-test credentials present during the earlier local raw-report generation.
8. Rerun the enterprise release gate and issue a new go/no-go decision.

Phase 3 must not begin until these Phase 2 controlled-pilot trust gates pass.

10. Rollback and Emergency Behavior

- Keep all tenant packages below ACTIVE_INTERNAL.
- The current certified package remains inactive and can be suspended without changing global definitions.
- If an active internal package later develops blocking drift, reconciliation suspends it through the canonical command.

- The kill switch remains an independent emergency deny.
- The additive migration should remain in place; rollback is application-level because removing idempotency evidence would reduce assurance.
- Immutable Workflow Assurance findings and release audit evidence must not be deleted.

11. Execution Notes

An existing local development process held the Prisma query-engine file open. A temporary generated-client directory was used to complete local validation without replacing the locked same-version engine. The temporary schema and client directory were verified as workspace-local and removed after use. No workaround artifact remains.

The dedicated patch tool also failed repeatedly because of a Windows sandbox helper error. Narrow assertion-guarded edits were used as the fallback and were then validated by lint, typecheck, tests, runtime gates, PostgreSQL smoke, browser certification, and `git diff --check`.

12. Final Decision

The Phase 2A release-control hardening slice is technically complete and verified locally. Stoquify now has materially stronger replay safety, tenant isolation, drift response, and evidence hygiene.

The overall Phase 2 controlled pilot is still **BLOCKED** on real governance, deployment, alert acknowledgement, clean-commit certification, credential rotation, and the complete negative/degradation matrix.

Current authorized status: READY FOR REMEDIATION VERIFICATION **Not authorized:** internal activation or Phase 3 execution **Active numbered skill:** 016-aqstoqflow-ai-copilot-guardrails **Next numbered skill after remaining evidence is supplied:** 017-aqstoqflow-enterprise-release-gate